



GhostChat: Complete Setup & Deployment Guide



This guide will take you from a single file on your computer to a fully live, globally accessible, end-to-end encrypted chat application. We will use **Firebase** for the free database and **Cloudflare Pages** for free, blazing-fast hosting.

Phase 1: Prepare Your Local File

1. Create a new folder on your computer desktop and name it `ghostchat`.
2. Open your preferred text editor (like Notepad, TextEdit, or VS Code).
3. Copy the complete `index.html` code from our editor on the right.
4. Paste the code into your text editor and save the file inside the `ghostchat` folder as exactly: `index.html`.

Phase 2: Set Up Firebase (Your Free Database)

Firebase will act as the lightning-fast relay for your encrypted messages.

A. Create the Project

1. Go to the [Firebase Console](#) and log in with your Google account.
2. Click **Add project**.
3. Name it something like `ghostchat-db` and click **Continue**.
4. Turn **OFF** Google Analytics (you don't need it for a private chat app), then click **Create project**. Wait for it to finish, then click **Continue**.

B. Create the Firestore Database

1. In the left-hand menu, click **Firestore Database**.
2. Click the **Create database** button.
3. Keep the location default, click **Next**.
4. Choose **Start in test mode** (this allows you to read/write for 30 days) and click **Enable**.
 - *Security Note: Because your app uses Military-grade AES-256 encryption in the browser, your data is safe even if the database is temporarily "open" in test mode.*

C. Get Your Configuration Keys

1. In the left menu, click the **Project Overview** gear icon  and select **Project settings**.
2. Scroll down to the "Your apps" section and click the **Web icon** (`</>`).
3. App nickname: `GhostChat Web`, click **Register app**.
4. You will see a block of code with a `firebaseConfig` object. **Keep this tab open**; you will need these keys in Phase 3.

Phase 3: Connect Your App to Firebase

Now we will tell your `index.html` file how to talk to your new database.

1. Open your `index.html` file in your text editor.
2. Scroll down to approximately line 150. You are looking for these two lines of code:

```
const [connectionMode, setConnectionMode] = useState("demo");
const [firebaseConfig, setFirebaseConfig] = useState(() => {
  const saved = localStorage.getItem("GHOST_FIREBASE_CONFIG");
  return saved ? JSON.parse(saved) : null;
});
```

3. **Replace** those lines entirely with your configuration from Firebase. It should look exactly like this (but with your actual keys):

```
const [connectionMode, setConnectionMode] = useState("firebase");
const [firebaseConfig, setFirebaseConfig] = useState({
  apiKey: "YOUR_API_KEY_FROM_FIREBASE",
  authDomain: "YOUR_PROJECT_ID.firebaseio.com",
  projectId: "YOUR_PROJECT_ID",
  storageBucket: "YOUR_PROJECT_ID.appspot.com",
  messagingSenderId: "YOUR_SENDER_ID",
  appId: "YOUR_APP_ID"
});
```

4. Save the `index.html` file.

(Your app is now fully configured! Anyone who opens this file will instantly connect to your live database).

Phase 4: Deploy to Cloudflare Pages (Free Hosting)

Now we will put the folder on the internet so you can access it from your phone.

1. Go to [Cloudflare Pages](#) and click **Sign Up** (or Log In).
2. On the left sidebar, click **Workers & Pages**.
3. Click the blue **Create** button, then select the **Pages** tab at the top.
4. Select **Upload assets** (Direct Upload).
5. Name your project (e.g., `my-ghost-chat`). Click **Create project**.
6. You will see an upload box. Drag and drop your entire `ghostchat` folder into this box (or click to browse and select the folder).
7. Once the `index.html` file shows as uploaded, click **Deploy site**.

Phase 5: You're Live! 🎉

Cloudflare will give you a custom link that looks something like `https://my-ghost-chat.pages.dev` .

How to use your new app:

1. Open your Cloudflare link on your Android or iPhone.
2. Type in your name and tap **Create Secure Room**.
3. Tap the **Share** button in the top right corner.
4. Send that link to your friend (via WhatsApp, SMS, Signal, etc.).

5. As soon as they click it, their browser will generate the matching decryption keys, and you can chat securely in real-time!

Troubleshooting:

- *Messages aren't syncing?* Double-check that you copied the Firebase Config correctly without missing any quotation marks, and that your database was created in "Test Mode".